

7 Carbon dioxide, methane and nitrous oxide are greenhouse gases.

Nitrous oxide is produced by bacteria in the soil.

- (a) State **one** anthropogenic source of carbon dioxide and **one** different anthropogenic source of methane.

(1)

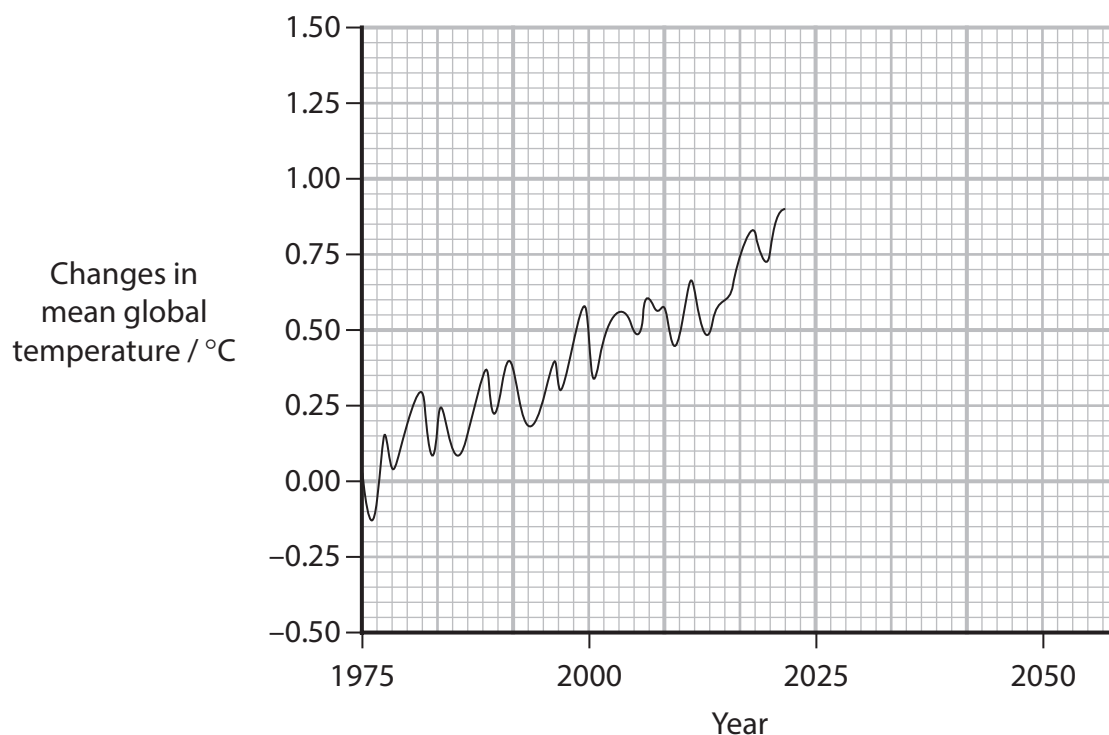
Carbon dioxide

Methane

- (b) Describe how greenhouse gases cause global warming.

(3)

(c) The graph shows the changes in mean global temperature from 1975 to 2023.



Predict the change in the mean global temperature in 2050 from the data in this graph.

(1)

Answer °C

(d) The greenhouse gases have different warming potentials (WP).

Over a 100-year timescale:

- one tonne of methane would generate 28 times the WP of one tonne of carbon dioxide
- one tonne of nitrous oxide would generate 265 times the WP of one tonne of carbon dioxide.

(i) Calculate the ratio of the WP of nitrous oxide to the WP of methane.

Give your answer to **two** significant figures.

(1)

Answer :1

- (ii) Carbon dioxide can last in the atmosphere for thousands of years. Methane lasts for about 12 years and nitrous oxide for about 121 years.

Discuss the implications of this data and the WPs when scientists are deciding how to tackle climate change.

(3)

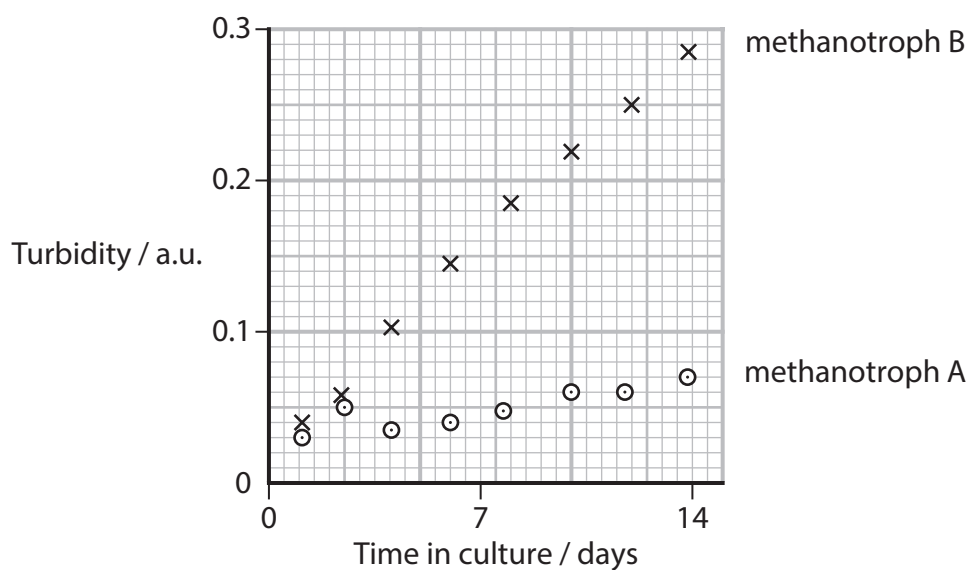
- (e) Methanotrophs are a group of bacteria that can absorb methane and use it as a source of energy and as a material to convert into biomass.

Most methanotrophs grow best in methane concentrations of 5 000 a.u. to 10 000 a.u. This is greater than the methane concentration in the atmosphere.

Scientists wanted to identify the methanotrophs that could grow at lower methane concentrations, as these could help reduce the levels of methane in the atmosphere.

- (i) The graph shows the growth curve for two types of methanotrophs grown in a methane concentration of 500 a.u.

The increase in turbidity of the culture was used to assess the growth of the methanotrophs.



Explain how this graph can be used to compare the growth rate of these two types of methanotrophs.

(2)

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(ii) Suggest why low methane concentrations are not able to support the growth of some methanotrophs, such as methanotroph A.

(1)

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(iii) Suggest why the scientists analysed the cultures of methanotroph B for the production of nitrous oxide and carbon dioxide.

(2)

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(Total for Question 7 = 14 marks)